

AI Predictive Maintenance System

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ABSTRACT

This project presents an AI-based Predictive Maintenance System using Arduino UNO and a temperature sensor for monitoring machine conditions. The system continuously measures the temperature of agricultural or industrial equipment using an LM35 sensor. When the temperature exceeds a predefined safe limit, a buzzer alert is activated to warn the operator about possible overheating or machine failure.

The main objective of this project is to reduce unexpected machine breakdowns and improve equipment safety through early fault detection. The system is simple, low-cost, and suitable for small-scale agricultural and industrial applications. It can help farmers and machine operators perform preventive maintenance before serious damage occurs.

This project also provides a basic foundation for future improvements such as IoT monitoring, cloud data storage, and AI-based fault prediction systems.

INTRODUCTION

Predictive maintenance is used to detect machine problems before failure occurs. This project uses an Arduino UNO, LM35 temperature sensor, and buzzer to monitor machine temperature. When the temperature exceeds the safe limit, the buzzer turns ON and gives an alert. This system helps reduce machine damage, improve safety, and support preventive maintenance in agricultural and industrial applications.

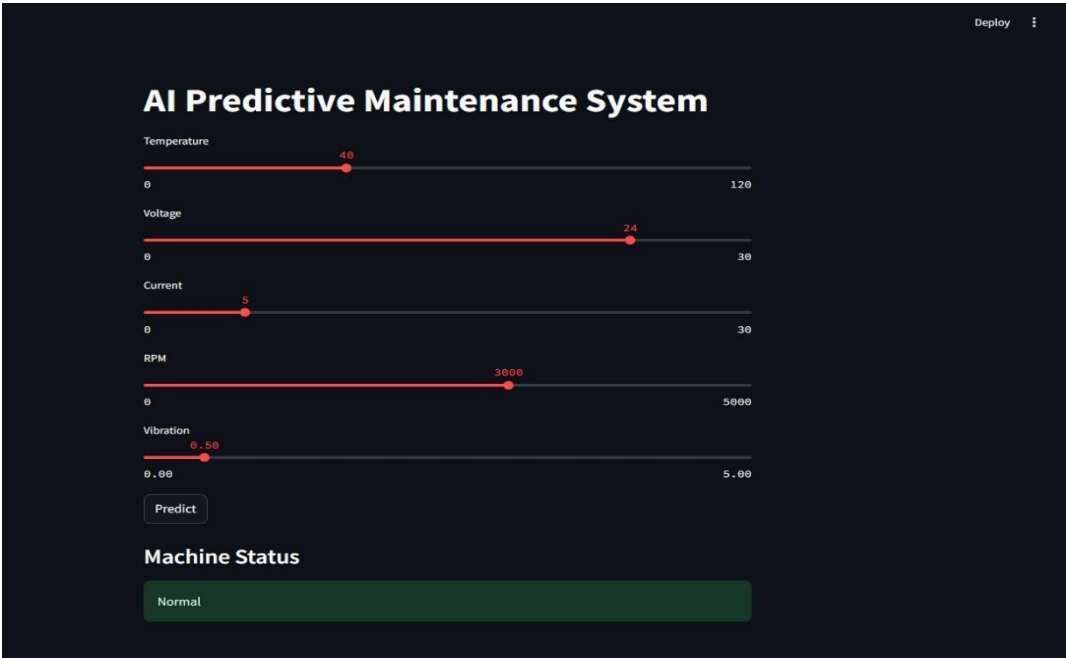
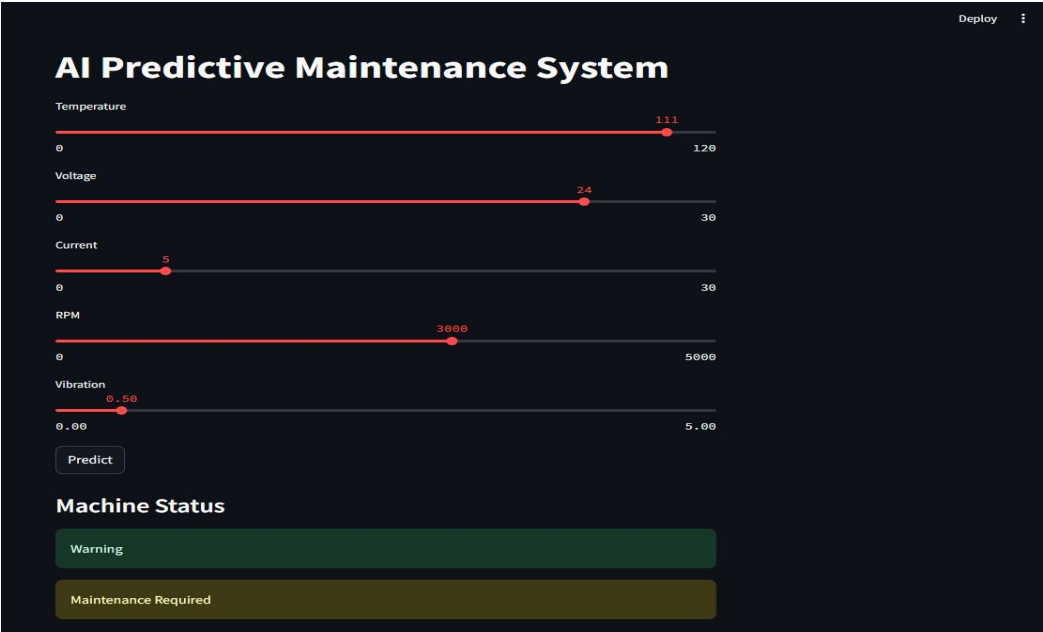
OBJECTIVES

- To monitor machine temperature
- To prevent overheating
- To provide early warning alerts
- To improve maintenance efficiency

COMPONENTS USED

S.No	Component
1	Arduino UNO
2	LM35 Sensor
3	Buzzer
4	Breadboard
5	Jumper Wires

Circuit Diagram



Working Principle

```
void loop() {  
    int tempValue = analogRead(tempPin);  
    float temperature = tempValue * 0.488;  
  
    if (temperature > 70) {  
        digitalWrite(buzzer, HIGH);  
    }  
    else {  
        digitalWrite(buzzer, LOW);  
    }  
  
    delay(1000);  
}
```

APPLICATIONS

- Agricultural machinery monitoring
- Motor temperature monitoring
- Preventive maintenance systems
- Industrial equipment safety

RESULT

The system successfully detects high temperature conditions and activates the buzzer alert system.

FUTURE SCOPE

- IoT integration
- Mobile app monitoring
- AI-based fault prediction
- Cloud data storage